Kashish Lalwani | New York, USA

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PROFESSIONAL SUMMARY

Applied Data Scientist & GenAI Engineer building evaluation-first LLM systems, production RAG pipelines, and scalable ML infrastructure. Proven experience designing end-to-end workflows (data → modeling → evaluation → deployment) at scale, processing 1M+ records and 100K+ queries while optimizing latency, throughput, and model reliability. Specialized in retrieval evaluation, bias diagnostics, and reproducible experimentation for robust, production-grade AI systems.

EDUCATION

Stony Brook University – *Master of Science in Data Science* **Expected May 2026**

Shivaji University – *B.Tech in Computer Science (AI & ML)* **Aug 2020 – May 2024**

EXPERIENCE

Data Scientist, Stony Brook University, USA ***(Retrieval Systems & ML Evaluation)*** **Aug 2025 – Dec 2025**

- Built SQL/Python pipelines with feature engineering processing 1M+ records and 100K+ queries for retrieval evaluation
- Designed validation and statistical testing workflows improving data quality, reproducibility, and model reliability metrics
- Developed ETL pipelines improving ingestion, transformation, and consistency across large-scale data workflows systems
- Performed statistical and root-cause analysis identifying semantic vs ranking instability and performance risks across pipelines
- Built monitoring workflows tracking latency, throughput, and execution metrics improving system scalability performance
- Optimized SQL queries and pipeline execution improving throughput efficiency and reducing latency across workflows systems
- Designed data models improving query performance and enabling scalable access across analytical retrieval systems
- Designed True-PIR evaluation metrics and bias diagnostics frameworks enabling systematic analysis of retrieval robustness

Founding AI Engineer, Versaunt (Aayats), Inc., USA ***(ML Systems, API & Data Pipelines)*** **May 2025 – Aug 2025**

- Built FastAPI backend services supporting 10K+ users and 500K+ queries for scalable AI system deployment systems
- Developed RAG pipelines using LLMs, embeddings, and vector search improving retrieval relevance and response accuracy metrics
- Designed A/B testing and experiment reproducibility frameworks processing 50K+ daily events for optimization workflows
- Engineered real-time data pipelines improving data consistency, analytics efficiency, and scalability across systems environments
- Implemented REST API services enabling integration of ML pipelines across backend workflows and microservices architecture
- Optimized inference workflows improving latency, throughput, and inference efficiency across production ML environments

Data Analyst Intern, Saroj Group of Companies, ***IN (Analytics & Automation)*** **Jan 2024 – July 2024**

- Applied statistical modeling on 100K+ records reducing defects by 40% and improving operational efficiency performance
- Performed financial analysis and forecasting improving profit margin by 12% through cost optimization strategies initiatives
- Built KPI dashboards using SQL, Excel, and Tableau enabling data visualization and improved decision-making processes
- Automated reporting pipelines reducing 25+ hours per month and improving reporting speed and data consistency accuracy

PROJECTS

Search Perturbation Robustness & Bias (PIR / PB) | *Python, Retrieval Evaluation, Statistical Testing*

- Built reproducible retrieval evaluation system quantifying ranking robustness under semantic-preserving query perturbations scenarios. Designed True-PIR metrics and bias diagnostics benchmarking cross-engine stability across search systems environments

RAG Financial Intelligence Platform ***(Confidential: Gov. of India)*** | *Python, LLMs, LangChain, Vector Search, ETL*

- Built end-to-end RAG pipelines using LLMs, embeddings, and vector search for scalable financial analysis workflows systems. Optimized retrieval latency and inference flow improving response speed by 60% and decision efficiency by 40% performance

Scalable Retrieval Benchmarking System | *Python, FAISS, Vector Embeddings, NumPy*

- Built FAISS-based indexing pipelines simulating 10K–50K workloads for retrieval evaluation and benchmarking analysis. Profiled latency, throughput, and indexing efficiency identifying bottlenecks and improving system scalability performance

Structured Clinical Data Pipeline (HackRare) | *Python, NLP, spaCy, negspacy, Data Modeling*

- Built NLP pipelines using spaCy and negspacy converting clinical notes into structured HPO and GA4GH outputs datasets

AI Portfolio Risk & Forecasting Model | *Python, Pandas, Statistical Modeling, Optimization*

- Built predictive models using time-series forecasting and optimization enabling financial risk analysis workflows systems

Formula 1 Performance Analytics | *Python, XGBoost, Regression, Clustering*

- Built regression and clustering models on 150K+ records for prediction, segmentation, and statistical analysis tasks

SKILLS

Programming & Databases: Python, Java, SQL, C, C++, PostgreSQL, MySQL, BigQuery

ML / AI & GenAI: Machine Learning, NLP, LLMs, RAG, Embeddings, Vector Search, FAISS, LangChain, Feature Engineering, Model Evaluation, Cross-Validation, Statistical Testing, Predictive Modeling, Time-Series Forecasting, Optimization, Model Reliability, Semantic Retrieval, Model Interpretability

Systems & Production ML: Data Pipelines, ETL/ELT, Data Modeling, Experiment Reproducibility, A/B Testing, Inference Optimization, Production Monitoring, Latency & Throughput Optimization, System Stability

Backend & Infrastructure: FastAPI, REST APIs, Microservices, Kafka (event-driven systems), Docker, CI/CD (Jenkins), Airflow, AWS

LEADERSHIP

"Best Outgoing Student"| Student Council Secretary (3×) | Inter - Zonal Chess Champion | Rotaract Leadership (District Secretary)